Appendix

Compliance in regulatory gray areas: The case of the organic seed standard

Survey sample

1. Data collection method

- 2010: Data were collected by convenience sample only, shared by 23 organizations, including non-profits and foundations, organic certifiers, and departments of agriculture. Surveys were distributed primarily over email, but paper, phone, and in-person surveys were conducted at the request of the grower. We received a total of 1,027 responses (712 web-based, 315 paper-based), 539 of which included complete enough data for our analysis.
- 2015: Data were collected by both random and convenience sampling. For the random sample, 800 certified organic producers were randomly selected from the approximately 12,000 organic producers that use seed, recorded in the NOP database. To encourage responsiveness, three follow-up emails were sent following an initial email, paper copies were mailed on request, and individual phone calls were made to follow-up. From these 800, 197 (25%) responded (114 web-based, 83 paper-based) and 122 (16%) were complete enough to use in our analyses. For the convenience sample, surveys were shared by 41 organizations, primarily over email. We received a total of 1,148 responses from the convenience sample, 648 of which included complete enough data for our analysis. In total we had 770 usable responses.
- 2020: Data were collected by both random and convenience sampling using methods similar to 2015. For the random sample, 2,000 certified organic producers were randomly selected from the the NOP database and given the option to complete the survey over email or paper. After an initial email and postcard invitation, growers received follow-up contacts by email, postcard, and phone, and hard copies of the survey were sent. Survey distribution was managed by Washington State University's Social and Economic Sciences Research Center (SESRC) and distributed alongside the Organic Farming Research Foundation's (OFRF) National Organic Research Agenda. From these 2,000 surveys, 448 (22%) responded (206 web-based and 242 paper-based) and 181 (10%) were complete enough to use in our analyses. For the convenience sample, surveys were distributed by SESRC to all remaining certified organic operations over email, with postcard reminders. The survey was also advertised by Organic Seed Alliance (OSA) and OFRF. We received a total of 611 responses from the convenience sample, 229 of which included complete enough data for our analysis. In total we had 410 usable responses.

The breakdown between random and convenience sampling used in our analysis is summarized in Table S1. We consider the effect of sampling method, as well as survey distributor, through models in Section 2.

Table S1: Convenience and random samples, usable responses

Year	Total usable responses	Convenience sample responses	Random sample responses
2010	539	539 (100 %)	0 (0 %)
2015	770	648 (84 %)	122 (16 %)
2020	410	229 (56 %)	181 (44 %)

2. Checking for biases induced by data collection methods

In order to test how our data collection methods may bias the sample, we consider the effect of the sampling approach and survey distributor in our model. First we model the effect of sampling method with our 2015 and 2020 survey data (the survey periods where we had both sampling approaches) by including a term ("Random data collection (vs. convenience)") for responses that were gathered by random sampling compared to convenience sampling (Table 2, Model 1). We see a small and insignificant effect for sampling method, and negligible changes in the results between the models run with and without the sampling consideration (Table 2, Model 2), suggesting no significant effect on results based on sampling approach.

Regarding concerns of survey distributor (i.e. that the distribution by Organic Seed Alliance (OSA) elicited responses from a select type of organic farmer), we model this effect with our 2010 and 2015 survey data (the only survey periods that identified specific distributors) by including a term ("Distributor: OSA (vs. Other)") for responses that were gathered through OSA channels compared to all others (Table 2, Model 3). We see a small and insignificant effect for distributor, and negligible changes in the results between the models run with and without the distributor consideration (Table 2, Model 4), suggesting no significant effect on results based on survey distributor.

Table S2: Beta regression model results including sampling and distributor effects

	Proportion of organic land planted with organic seed			
	Including sampling method (2015, 2020)	Without sampling method (2015, 2020)	Including survey distributor (2010, 2015)	Without survey distributor (2010, 2015)
	(1)	(2)	(3)	(4)
Ability	0.684***	0.686***	0.624***	0.616***
	(0.146)	(0.146)	(0.149)	(0.149)
Duty	0.272***	0.271***	0.231***	0.229***
	(0.048)	(0.047)	(0.043)	(0.043)
Ability x Duty	−0.127***	−0.127***	−0.128***	−0.125***
	(0.041)	(0.041)	(0.042)	(0.042)
Certifier request	−0.038	−0.036	−0.185**	−0.182**
	(0.079)	(0.079)	(0.075)	(0.075)
Farm size (log acres)	−0.121***	−0.121***	−0.127***	−0.125***
	(0.025)	(0.025)	(0.024)	(0.023)
Crop diversity	0.110	0.110	0.170***	0.171***
	(0.070)	(0.069)	(0.064)	(0.064)
Seed saved or traded (%)	0.052	0.052	0.079**	0.078**
	(0.039)	(0.039)	(0.035)	(0.035)
Years certified organic	0.023	0.023	0.044	0.043
	(0.036)	(0.036)	(0.039)	(0.039)
Forage crops (vs. Field)	−0.851***	−0.849***	−0.909***	−0.910***
	(0.112)	(0.112)	(0.101)	(0.101)
Vegetable crops (vs. Field)	−0.532***	−0.534***	−0.603***	−0.610***
	(0.127)	(0.127)	(0.120)	(0.120)
Western region (vs. North Central)	−0.332***	−0.331***	−0.320***	−0.367***
	(0.103)	(0.103)	(0.103)	(0.099)
Northeastern region (vs. North Central)	−0.348***	−0.343***	−0.220*	−0.279**
	(0.108)	(0.107)	(0.120)	(0.115)
Southern region (vs. North Central)	−0.070	−0.071	−0.190	−0.222*
	(0.139)	(0.139)	(0.134)	(0.132)
Year: 2020 (vs. 2015)	0.037	0.047		
	(0.084)	(0.080)		
Random data collection (vs. convenience)	0.036			
	(0.091)			
Year: 2010 (vs. 2015)			−0.257***	−0.250***
			(0.080)	(0.080)
Distributor: OSA (vs. Other)			−0.139*	
			(0.083)	
Constant	0.134	0.144	0.403**	0.378**
	(0.185)	(0.183)	(0.168)	(0.168)
BIC	-3947	-3954	-4028	-4032
Observations	1,180	1,180	1,309	1,309
R^2	0.239	0.239	0.237	0.235

Note: *p<0.1; **p<0.05; ***p<0.01
R-squared reported is pseudo R-squared

3. Sample representativeness

While the mix of sampling methods creates a challenge for determining the sample's representativeness of the population, we aim to demonstrate that our sample is representative by comparing the geographic distribution of operations (Table S3) and farm size distribution (Figure S1) between our sample and USDA Census data.

We first use USDA Census data of organic farmers from 2011, 2016, and 2021 to align with our 2010, 2015, and 2020 survey periods. These data are made available by the National Agricultural Statistics Service Quick Stats API. Across the three survey periods, the number of respondents represented approximately 6.3%, 5.9%, and 2.5% of the total number of organic farmers identified in the USDA Census in the related years. In Table S3 we compare the number of operations in our sample (n) to the population (N) by year and region. For each census period we calculated the standard deviation of the percentage of operations represented, which we used to identify regions that are over or under-represented. Over-represented year-region combinations are signaled by italic text formatting, while under-represented year-region combinations are signaled by bold text formatting. Only in 2010 do we see significant variation in representation across regions, with the North Central and Southern regions over-represented and the Northeast under-represented.

Table S3: Sample representativeness compared to USDA Censuses (2011, 2016, 2021)

Year	Region	Operations in sample (n)	Operations in USDA census (N)	Percent of operations represented (n/N)
2010	North Central	302	2986	*10.1*
2010	Northeast	49	1897	**2.6**
2010	South	51	540	*9.4*
2010	West	137	3296	4.2
2015	North Central	378	4689	8.1
2015	Northeast	137	3246	4.2
2015	South	72	1096	6.6
2015	West	183	4529	4
2020	North Central	175	5824	3
2020	Northeast	94	3888	2.4
2020	South	37	1209	3.1
2020	West	104	5273	2

Note:
Italic formatting sigals over-representation and bold formatting signals under-representation

Next, to understand farm size representativeness, we look at USDA Census organic tabulations, reported for as the 2012, 2017, and 2022 census tabulations.

In Figure S1 we compare these USDA Census data against our survey sample. These comparisons show a consistent over-representation of the smallest farms (1-9 acres), especially in the North Central and South, and under-representation of the largest farms. Still, our sample captures a distribution of organic farm sizes that includes medium and large farms, allowing us to understand the effect of farm size in our analysis. Additionally, an important consideration is that we calculate farm size as reported acreage planted to a grower's main crop, focusing on three major crop categories (field, forage, and vegetable). However, this approach likely underestimates acreage compared to USDA Census data, which reports all certified land. Furthermore these data are not crop specific and so USDA data also includes non-seed crop operations (i.e. orchards).

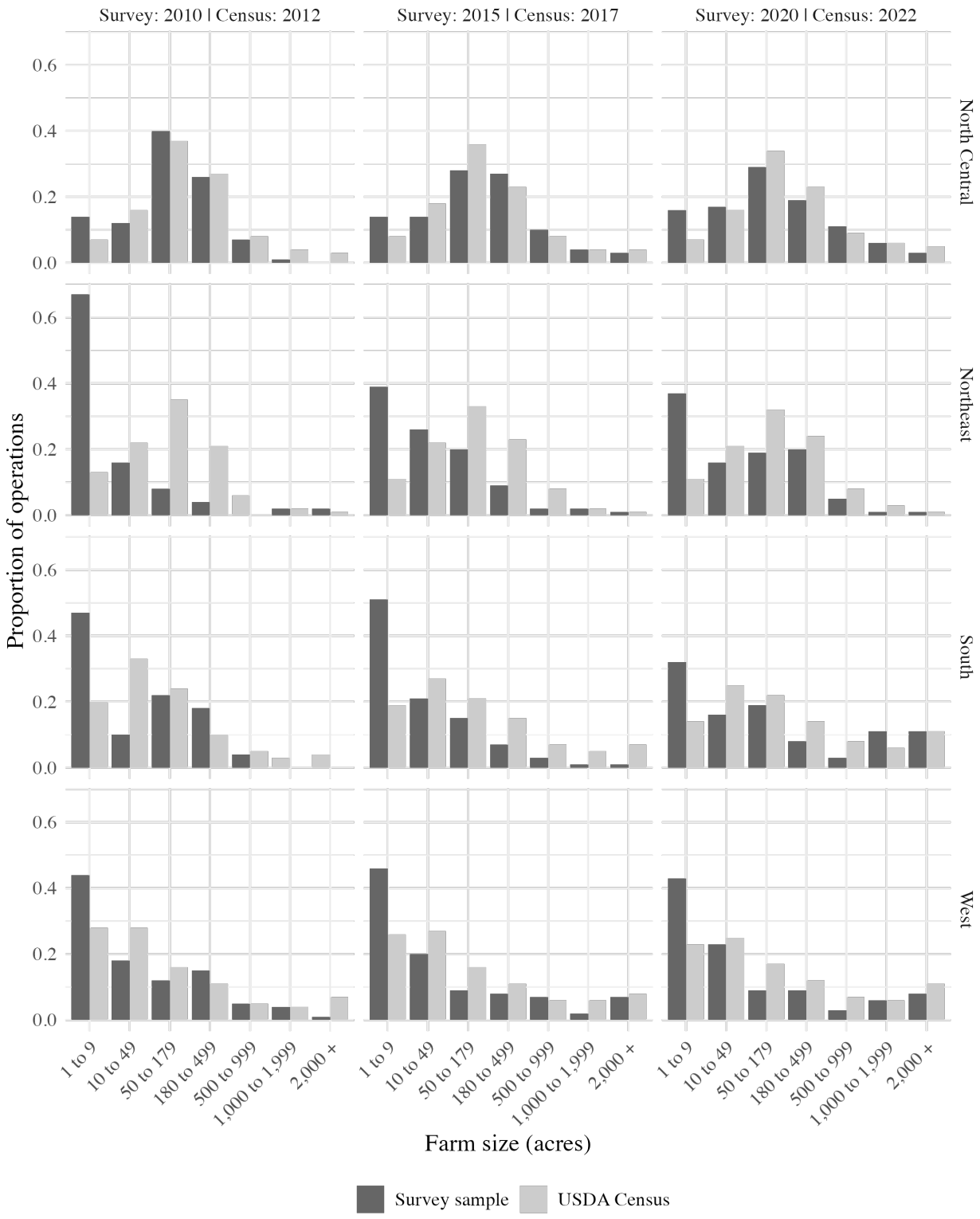


Figure S1: Sample farm size distribution compared to USDA Census data

Survey variation

4. Survey questions for key variables across survey periods

Across the three time periods that data were collected there were some changes to question wording. Here we report full survey question text for each year and explain any necessary transformations.

Acreage planted to organic seed [vegetable crop example]

2010:
Q3a. How many organic acres in a given season do you have in [annual vegetables and fruits]?
Q3b. What approximate percentage of the total acreage of the above crops is planted with certified organic seed?

2015:
Q3. Last year, did you have any organic acres in annual vegetables? (Y/N)
The following two questions are conditional on the previous question:
Q4. How many organic acres did you have in [annual vegetables] last year?
Q5. Last year, what approximate percentage of total acreage of [annual vegetables] was planted with certified organic seed?

2020:
Q5. How many certified organic acres did you have in [annual vegetable crops] last year?
The following question is conditional on the previous question:
Q5A. Last year, approximately what percentage of your certified organic [annual vegetable crops] were planted with certified organic seed?

Duty

2010:
Q17. In your opinion, to what degree is using organic seed in organic systems important to the integrity of organic food production? (very important/high priority, important/medium priority, not very important/low priority, not at all important/not a priority, uncertain)

Q18. In your opinion, are varieties bred for organic system management important to the overall success of organic agriculture? (very important/high priority, important/medium priority, not very important/low priority, not at all important/not a priority, uncertain)

2015 / 2020:
Q27. / Q30. Please indicate the extent to which you disagree or agree with the following statements. (strongly disagree, disagree, neutral, agree, strongly agree)
- Organic seed is important to the integrity of organic food production
- Varieties bred for organic production are important to the overall success of organic agriculture

Transformation: The 'uncertain' category from 2010 was re-categorized as 'neutral' to align with the scale from the 2015 and 2020 surveys.

Ability

2010:
Q10. Using a scale of 1 = "not a factor" to 5 "significant factor" please share the reasons you have not purchased organic seed in the past. Please circle one response per reason below. (1 = not a factor; 2 = slight factor; 3 = moderate factor; 4 = more than moderate; 5 = significant factor)

2015:
Q23. For each reason listed below, how much of a reason was it for you to *not purchase* organic seed over the last three years (2011-2013)? (not a reason, slight reason, moderate reason, significant reason)

2020:
Q29. Over the last three years (2017-2019), how much were each of the following a factor in your decision NOT to purchase organic seed? (not a factor, slight factor, moderate factor, significant factor)

Transformation: 5-level categories from 2010 were converted into 4 categories, where those with 'more than moderate' responses in 2010 we joined with the 'significant' grouping to align with the 2015 and 2020 surveys.

5. Exploring variation in acreage planted to organic seed survey question

The survey question related to *acreage planted to organic seed*, our key dependent variable, is asked with different levels of precision across years. This change in precision requires further investigation, particularly to determine how respondents with split operations (i.e. farms in which part of their acreage is certified organic and another part is not) understand and respond to the question.

The question was intended (and interpreted in the paper) as it was posed in 2020: *Last year, approximately what percentage of your certified organic [vegetable / field / forage] crops were planted with certified organic seed?*. However, previous years omit the specific pointer to **certified organic** crops, and instead ask *Last year, what approximate percentage of total acreage of [vegetable / field / forage] was planted with certified organic seed?* (2015) or *What approximate percentage of the total acreage of the above crops is planted with certified organic seed?* (2010). These questions follow / are conditional on the previous question which is specifically about the organic acreage they have under production in a given crop. We expect that respondents will take the cue from the previous question and therefore respond about their organic seed use based only on their organic acreage.

To understand whether this question was interpreted as we expected, we carry out two explorations of the data: 1) Disaggregating results by year and 2) Identifying split operations with responses across multiple surveys.

Models disaggregated by year

The survey question related to acreage planted to organic seed is most precise in 2020, and so we would expect that the 2020 results should represent our overall expectations from the model. And indeed in Table S4 we observe results in 2020 that are consistent with the overall model results, suggesting that the questions are interpreted as we expect. Further, we can include a dummy variable for split operations in the 2020 data. We observe that that the effects of our key variables of interest do not change across years. Furthermore, when question wording is least precise in 2015, we would expect that the effect of farm size would change if the question was misinterpreted by split operations. Specifically, we would expect an increase in the effect size of farm size (i.e. a large, negative effect). This is because if split operations (which are on average, almost twice as large as fully organic operations according to our 2020 data) had been interpreting the question about organic seed use to apply to all of their acreage, their reported percentage planted to organic seed would be much lower. However, the effect of farm size in 2015 is similar to other years.

Split operations with multi-year responses

To further explore the consequence of question imprecision on split operations, we tried to find respondents who we knew had split operations (based a survey question only included in the 2020 survey) and who also responded to previous surveys. However, because the surveys were not intended to be identifiable we had only limited methods for tracking repeated respondents (i.e. email addresses provided for follow-up contact, verified by location and crop matches). We found only one identifiable, repeat respondent who is known to have a split operation. We compared their answers between the 2015 and 2020 survey to determine whether they interpreted the questions differently.

In 2020, when asked, *Last year, approximately what percentage of your certified organic [vegetable / field / forage] crops were planted with certified organic seed?*, the respondent replied 100% for both crop types that they planted: field crops and forage crops. In 2015, when asked, *Last year, what approximate percentage of total acreage of [vegetable / field / forage] crop was planted with certified organic seed?*, the respondent also replied 100% for both crop types that they planted: field crops and forage crops. The consistent response across surveys, despite less precise question wording in 2015, suggests that this respondent took cues from the previous questions and understand the acreage in focus to be organic acreage.

Table S4: Model results by year

	Proportion of production acreage planted with organic seed			
	2010	2015	2020	2020 incl. split
Ability	0.172***	0.213***	0.375***	0.383***
	(0.056)	(0.048)	(0.064)	(0.064)
Duty	0.162***	0.238***	0.156**	0.154**
	(0.057)	(0.047)	(0.066)	(0.065)
Ability x Duty	−0.099*	−0.112**	−0.104*	−0.110*
	(0.051)	(0.045)	(0.062)	(0.062)
Certifier request	−0.313***	−0.100	0.068	0.084
	(0.117)	(0.097)	(0.135)	(0.135)
Farm size (log acres)	−0.131***	−0.130***	−0.112**	−0.102**
	(0.039)	(0.029)	(0.046)	(0.046)
Crop diversity	0.177***	0.047	0.126*	0.123*
	(0.062)	(0.050)	(0.070)	(0.070)
Seed saved or traded (%)	0.180***	0.013	0.098	0.087
	(0.059)	(0.047)	(0.064)	(0.064)
Years certified organic	0.074	0.021	0.022	0.003
	(0.061)	(0.046)	(0.067)	(0.068)
Forage crops (vs. Field)	−0.703***	−1.082***	−0.478**	−0.476**
	(0.149)	(0.137)	(0.192)	(0.192)
Vegetable crops (vs. Field)	−0.503***	−0.767***	−0.120	−0.130
	(0.192)	(0.155)	(0.226)	(0.225)
Western region (vs. North Central)	−0.534***	−0.228*	−0.514***	−0.493***
	(0.153)	(0.128)	(0.173)	(0.173)
Northeastern region (vs. North Central)	−0.157	−0.319**	−0.389**	−0.378**
	(0.218)	(0.137)	(0.170)	(0.170)
Southern region (vs. North Central)	−0.427**	−0.068	−0.081	−0.076
	(0.205)	(0.172)	(0.237)	(0.237)
Split operation				−0.342*
				(0.206)
Constant	0.925***	1.310***	0.913***	0.946***
	(0.128)	(0.104)	(0.143)	(0.144)
BIC	-1068	-2342	-1019	-1016
Observations	539	770	410	410
R^2	0.183	0.272	0.229	0.236

Note: *p<0.1; **p<0.05; ***p<0.01
R-squared reported is pseudo R-squared

Model checks

6. Correlations of all ability factors (Table S5)

Table S5: Correlations among organic seed sourcing ability factors

Barrier	Availability	Quantity	Quality	Form	Price
Quantity	0.50				
Quality	0.64	0.42			
Form	0.21	0.22	0.23		
Price	0.36	0.31	0.39	0.34	
Buyer requirement	0.41	0.39	0.27	0.21	0.22

7. Dissagregating allowable and non-allowable factors (Table S6)

Table S6: Beta regression model results disaggregating allowable and non-allowable ability factors

	Proportion of production acreage planted with organic seed					
	(1)	(2)	(3)	(4)	(5)	(6)
Ability: Allowable	0.231***		0.200***	0.457***	0.199***	0.381***
	(0.032)		(0.034)	(0.130)	(0.034)	(0.140)
Ability: Non-allowable		0.182***	0.129***	0.124***	0.373***	0.299**
		(0.034)	(0.035)	(0.035)	(0.123)	(0.133)
Duty	0.219***	0.202***	0.199***	0.201***	0.188***	0.193***
	(0.039)	(0.039)	(0.039)	(0.039)	(0.039)	(0.039)
Ability: Allowable x Duty				−0.075**		−0.053
				(0.037)		(0.040)
Ability: Non-allowable x Duty					−0.074**	−0.053
					(0.036)	(0.039)
Certifier request	−0.123*	−0.141**	−0.085	−0.089	−0.089	−0.091
	(0.066)	(0.066)	(0.066)	(0.066)	(0.066)	(0.066)
Farm size (log acres)	−0.126***	−0.127***	−0.118***	−0.115***	−0.116***	−0.115***
	(0.021)	(0.021)	(0.021)	(0.021)	(0.021)	(0.021)
Crop diversity	0.174***	0.159***	0.179***	0.183***	0.181***	0.184***
	(0.057)	(0.057)	(0.057)	(0.057)	(0.057)	(0.057)
Seed saved or traded (%)	0.091***	0.097***	0.098***	0.098***	0.096***	0.096***
	(0.032)	(0.032)	(0.032)	(0.032)	(0.032)	(0.032)
Years certified organic	0.043	0.036	0.040	0.039	0.041	0.040
	(0.032)	(0.033)	(0.032)	(0.032)	(0.032)	(0.032)
Forage crops (vs. Field)	−0.799***	−0.825***	−0.815***	−0.818***	−0.817***	−0.818***
	(0.090)	(0.090)	(0.090)	(0.090)	(0.090)	(0.090)
Vegetable crops (vs. Field)	−0.510***	−0.553***	−0.490***	−0.476***	−0.475***	−0.469***
	(0.106)	(0.105)	(0.106)	(0.106)	(0.106)	(0.106)
Western (vs. N. Cntrl.)	−0.392***	−0.433***	−0.394***	−0.393***	−0.400***	−0.397***
	(0.086)	(0.086)	(0.086)	(0.086)	(0.086)	(0.086)
Northeastern (vs. N. Cntrl.)	−0.335***	−0.376***	−0.323***	−0.323***	−0.328***	−0.327***
	(0.095)	(0.095)	(0.095)	(0.095)	(0.095)	(0.095)
Southern (vs. N. Cntrl.)	−0.192*	−0.225*	−0.192*	−0.188	−0.196*	−0.192*
	(0.116)	(0.116)	(0.116)	(0.116)	(0.116)	(0.116)
Year: 2010 (vs. 2015)	−0.240***	−0.284***	−0.211***	−0.207***	−0.212***	−0.210***
	(0.079)	(0.078)	(0.079)	(0.079)	(0.079)	(0.079)
Year: 2020 (vs. 2015)	0.051	0.077	0.077	0.073	0.075	0.073
	(0.080)	(0.080)	(0.080)	(0.080)	(0.080)	(0.080)
Constant	0.308**	0.433***	0.330**	0.325**	0.378**	0.361**
	(0.152)	(0.152)	(0.152)	(0.151)	(0.154)	(0.153)
BIC	-5753	-5731	-5759	-5756	-5756	-5750
Observations	1,719	1,719	1,719	1,719	1,719	1,719
R^2	0.216	0.198	0.225	0.226	0.226	0.227

Note: *p<0.1; **p<0.05; ***p<0.01
R-squared reported is pseudo R-squared

8. Dissagregating all barriers (Table S7)

Table S7: Beta regression model results disaggregating all ability factors

	Proportion of production acreage planted with organic seed
Ability: Availability (Allowable)	0.173***
	(0.038)
Ability: Quality (Allowable)	0.116***
	(0.037)
Ability: Quantity (Allowable)	−0.001
	(0.036)
Ability: Form (Allowable)	−0.059*
	(0.033)
Ability: Price (Non-allowable)	0.052
	(0.034)
Ability: Buyer Requirement (Non-allowable)	0.117***
	(0.034)
Duty	0.185***
	(0.039)
Certifier request	−0.086
	(0.066)
Farm size (log acres)	−0.109***
	(0.021)
Crop diversity	0.173***
	(0.057)
Seed saved or traded (%)	0.095***
	(0.032)
Years certified organic	0.043
	(0.032)
Forage crops (vs. Field)	−0.809***
	(0.090)
Vegetable crops (vs. Field)	−0.468***
	(0.107)
Western region (vs. North Central)	−0.397***
	(0.087)
Northeastern region (vs. North Central)	−0.348***
	(0.096)
Southern region (vs. North Central)	−0.176
	(0.116)
Year: 2010 (vs. 2015)	−0.204***
	(0.079)
Year: 2020 (vs. 2015)	0.094
	(0.081)
Constant	0.366**
	(0.152)
BIC	-5748
Observations	1,719
R^2	0.236

Note: *p<0.1; **p<0.05; ***p<0.01
R-squared reported is pseudo R-squared

Observed data

9. Typeology table (Table S8)

Table S8: Descriptive summaries of observed duty-ability typologies

Ability	Duty: Low (<= Neutral)	Duty: High (> Neutral)
High (> None)	Avg. organic seed use: 81% Operations: 2% (n = 35)	Avg. organic seed use: 82% Operations: 23% (n = 389)
Low (None)	Avg. organic seed use: 40% Operations: 10% (n = 177)	Avg. organic seed use: 62% Operations: 65% (n = 1118)